

Wazuh File Integrity Monitoring (FIM) Project

Project Overview

This project focused on implementing and validating File Integrity Monitoring (FIM) using Wazuh in a home SOC environment. The objective was to configure Wazuh Syscheck to monitor a custom directory on a Windows Server 2019 endpoint and verify that file creation, modification, and deletion events were successfully detected and reported to the Wazuh Manager.

The project also involved troubleshooting a configuration issue that initially prevented File Integrity Monitoring events from being generated.

Lab Environment

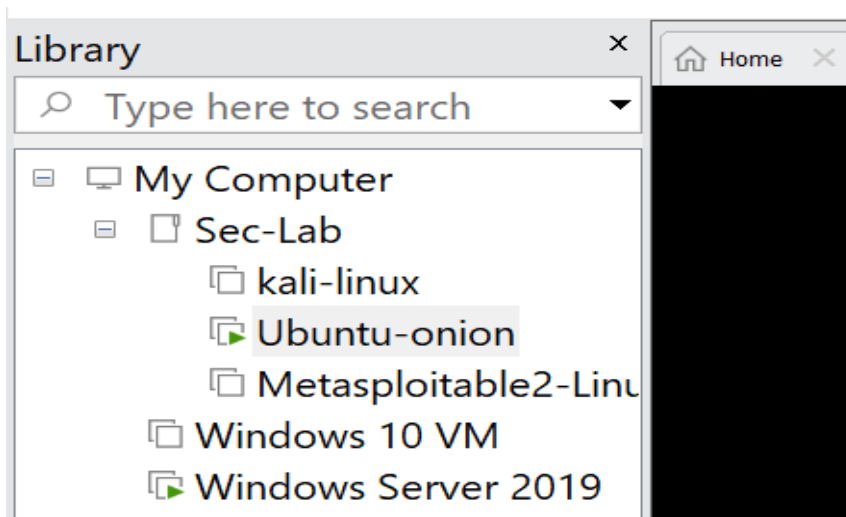


Figure 1: Virtual machines configured in the SOC home lab environment.

Lab Architecture

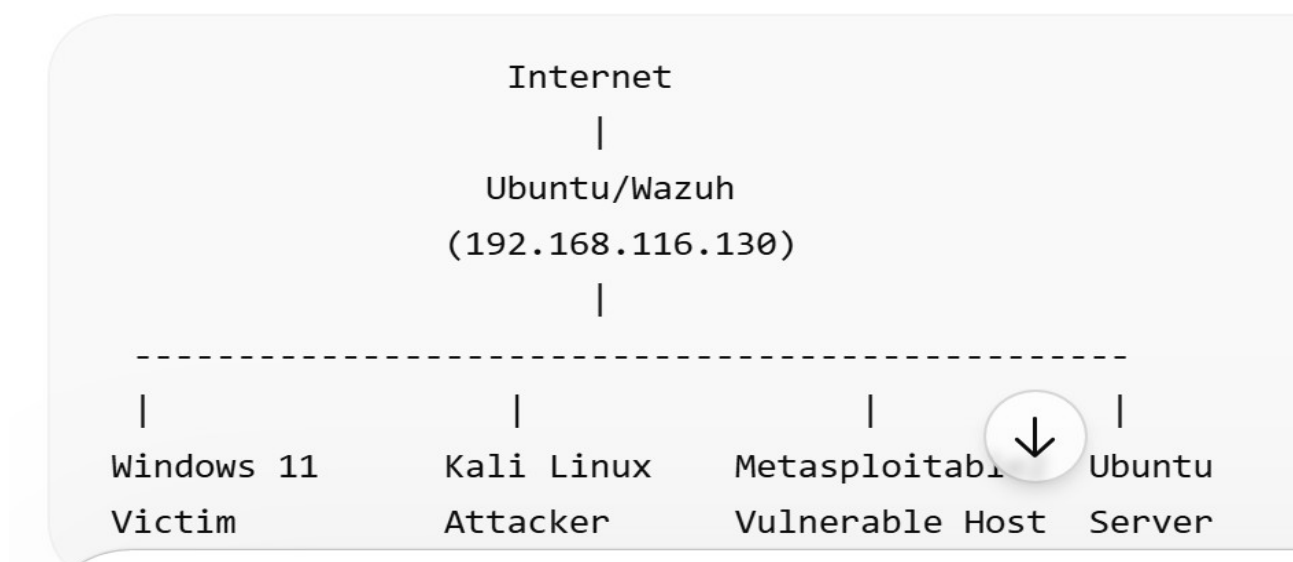


Figure 2: SOC Home Lab Architecture used for Wazuh File Integrity Monitoring (FIM) testing.

The RainhaStar Windows Server 2019 endpoint was monitored by the Wazuh Manager running on Ubuntu 24.04.

Wazuh Manager Status

- Operating System: Ubuntu 24.04 LTS
- Hostname: margate-VMware-Virtual-Platform
- IP Address: 192.168.116.130

Monitored Endpoints

Endpoint 1

- Hostname: Win11-Host
- Operating System: Windows 11
- Wazuh Agent ID: 001

Endpoint 2

- Hostname: RainhaStar
- Operating System: Windows Server 2019
- Wazuh Agent ID: 002

Hostname	Operating System	Agent ID
Win11-Host	Windows 11	001
RainhaStar	Windows Server 2019	002

Figure 3 Wazuh Agent Status

Project Objectives

The primary objectives of this project were:

- Configure Wazuh Syscheck on Windows Server 2019.
- Create a dedicated File Integrity Monitoring directory.
- Generate file creation events.
- Generate file modification events.
- Generate file deletion events.
- Verify alert generation within the Wazuh Dashboard. Investigate and troubleshoot configuration issues.
- Validate event transmission from the endpoint to the Wazuh Manager.

Initial Configuration

A test directory was created on the Windows Server endpoint:

New-Item -ItemType Directory -Path [C:\FIM-Test](#)

```
PS C:\Users\Administrator> New-Item -Path C:\FIM-Test\test3.txt -ItemType File

Directory: C:\FIM-Test

Mode                LastWriteTime         Length Name
----                -
-a-----         05.06.2026   12:09             0 test3.txt

PS C:\Users\Administrator>
```

Figure 4: Creation of File

The Wazuh agent configuration file was modified:

Location:

C:\Program Files (x86)\ossec-agent\ossec.conf

The following Syscheck configuration was added:

```
<directories realtime="yes">C:\FIM-Test</directories>
```

Troubleshooting Process

Problem Identified

- Agent active
- No Syscheck alerts
- alerts.json showed nothing

Although the Wazuh agent was connected and actively sending Windows Security events, no File Integrity Monitoring alerts were generated for the RainhaStar endpoint.

Root Cause Analysis

The Syscheck directory configuration was accidentally placed inside the synchronization block:

Incorrect configuration:

```

PS C:\Users\Administrator> Get-Content "C:\Program Files (x86)\ossec-agent\ossec.conf" | Select-String "FIM-Test" -Context 3,3

    <!-- Database synchronization settings -->
    <synchronization>
>    <directories realtime="yes">C:\FIM-Test</directories>
        <enabled>yes</enabled>
        <interval>5m</interval>
        <max_eps>10</max_eps>
    </synchronization>

PS C:\Users\Administrator>

```

Figure 5:

Because of this placement, Wazuh did not monitor the custom directory.

Corrective Action

The configuration was corrected by moving the directory entry outside the synchronization block and directly under the Syscheck section.

Correct configuration:

```

<synchronization>
    <enabled>yes</enabled>
    <interval>5m</interval>
    <max_eps>10</max_eps>
</synchronization>

<directories realtime="yes">C:\FIM-Test</directories>

</syscheck>

```

The Wazuh service was then restarted:

```
Restart-Service WazuhSvc
```

```
Get-Service WazuhSvc
```

Verification:

```

PS C:\Users\Administrator> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\Users\Administrator> Restart-Service WazuhSvc
PS C:\Users\Administrator> Get-Service WazuhSvc

Status   Name      DisplayName
-----
Running  WazuhSvc  Wazuh

```

Figure 6:

File Creation Test

Wazuh detection of file creation event (Rule 554).

```
New-Item -Path C:\FIM-Test\test3.txt -ItemType File
```

```
PS C:\Users\Administrator> New-Item -Path C:\FIM-Test\test3.txt -ItemType File

Directory: C:\FIM-Test

Mode                LastWriteTime         Length Name
----                -
-a----           05.06.2026   12:09              0 test3.txt

PS C:\Users\Administrator>
```

Figure 7:

```
sudo grep '"id":"002"' /var/ossec/logs/alerts/alerts.json | grep syscheck | tail
-20
```

Creation of test3.txt within the monitored directory.

```
...EAD_EA","WRITE_EA"]}]}, {"uid_after": "S-1-5-32-544", "gid_before": "S-1-5-32-544", "gid_after": "S-1-5-32-544", "uname_after": "Administrators", "gname_after": "SYSTEM", "mtime_before": "2026-06-04T21:39:55", "mtime_after": "2026-06-04T21:41:20", "changed_attributes": ["mtime"], "event": "modified"}, {"decoder": {"name": "syscheck_registry_key_modified"}, "location": "syscheck"}], [{"timestamp": "2026-06-05T12:09:31.574+0200", "rule": {"level": 5, "description": "File added to the system.", "id": "554", "firedtimes": 1, "mail": false, "groups": ["ossec", "syscheck", "syscheck_entry_added", "syscheck_file"], "pci_dss": ["11.5"], "gpg13": ["4.11"], "gdpr": ["II 5.1.f"], "hipaa": ["164.312.c.1", "164.312.c.2"], "nist_800_53": ["SI.7"], "tsc": ["PI1.4", "PI1.5", "CC6.1", "CC6.8", "CC7.2", "CC7.3"]}, "agent": {"id": "002", "name": "RainhaStar", "ip": "192.168.116.150"}, "manager": {"name": "margrate-VMware-Virtual-Platform", "id": "1780654171.1480975", "full_log": "File 'c:\\fim-test\\test3.txt' added\nMode: realtime\n", "syscheck": {"path": "c:\\fim-test\\test3.txt", "mode": "realtime", "size_after": "0", "win_perm_after": [{"name": "SYSTEM", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Administrators", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Users", "allowed": ["READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "READ_EA", "EXECUTE", "READ_ATTRIBUTES"]}]}, "uid_after": "S-1-5-32-544", "md5_after": "d41d8cd98f00b204e9800998ecf8427e", "sha1_after": "da39a3ee5e6b4b0d3255bfef95601890afd80709", "sha256_after": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "attrs_after": ["ARCHIVE"], "uname_after": "Administrators", "mtime_after": "2026-06-05T12:09:30", "event": "added"}, {"decoder": {"name": "syscheck_new_entry"}, "location": "syscheck"}]
margrate@margrate-VMware-Virtual-Platform:~$
```

Figure 8:

Detection

Wazuh generated the following alert:

Rule ID:

554

Description:

File added to the system.

Additional Details:

- Agent: RainhaStar (002)
- Monitoring Mode: Realtime
- Path: C:\FIM-Test\test3.txt

Wazuh detection of file creation event (Rule 554).

```
175 325233000 105520422 3430331001 ; detected [ "GENERIC_READ", "READ_CONTROL", "READ_DATA", "READ_EA", "WRITE_EA" ] }, "uid_after": "S-1-5-32-544", "gid_before": "S-1-5-32-544", "gid_after": "S-1-5-18", "uname_after": "Administrators", "gname_after": "SYSTEM", "mtime_before": "2026-06-04T21:39:55", "mtime_after": "2026-06-04T21:41:20", "changed_attributes": [ "mtime" ], "event": "modified", "decoder": { "name": "syscheck_registry_key_modified" }, "location": "syscheck" }
{ "timestamp": "2026-06-05T12:09:31.574+0200", "rule": { "level": 5, "description": "File added to the system.", "id": "554", "firedtimes": 1, "mail": false, "groups": [ "ossec", "syscheck", "syscheck_entry_added", "syscheck_file" ], "pci_dss": [ "11.5" ], "gpg13": [ "4.11" ], "gdpr": [ "II 5.1.f" ], "hipaa": [ "164.312.c.1", "164.312.c.2" ], "nist_800_53": [ "SI.7" ], "tsc": [ "PI1.4", "PI1.5", "CC6.1", "CC6.8", "CC7.2", "CC7.3" ] }, "agent": { "id": "002", "name": "RainhaStar", "ip": "192.168.116.150" }, "manager": { "name": "margrate-VMware-Virtual-Platform", "id": "1780654171.1480975", "full_log": "File 'c:\\fim-test\\test3.txt' added\\nMode: realtime\\n", "syscheck": { "path": "c:\\fim-test\\test3.txt", "mode": "realtime", "size_after": "0", "win_perm_after": { "name": "SYSTEM", "allowed": [ "DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES" ] }, { "name": "Administrators", "allowed": [ "DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES" ] }, { "name": "Users", "allowed": [ "READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "READ_EA", "EXECUTE", "READ_ATTRIBUTES" ] } } }, "uid_after": "S-1-5-32-544", "md5_after": "d41d8cd98f00b204e9800998ecf8427e", "sha1_after": "da39a3ee5e6b4b0d3255bfe95601890afd80709", "sha256_after": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "attrs_after": [ "ARCHIVE" ], "uname_after": "Administrators", "mtime_after": "2026-06-05T12:09:30", "event": "added", "decoder": { "name": "syscheck_new_entry" }, "location": "syscheck" }
margrate@margrate-VMware-Virtual-Platform:~$
```

Figure 9:

File Modification Test

Action

```
PS C:\Users\Administrator> Add-Content C:\FIM-Test\test3.txt "This is a modification test"
PS C:\Users\Administrator> Add-Content C:\FIM-Test\test3.txt "This is a modification test"
PS C:\Users\Administrator> Remove-Item C:\FIM-Test\test3.txt
Remove-Item C:\FIM-Test\test3.txt
```

Figure 10:

The file was modified:

```
Add-Content C:\FIM-Test\test3.txt "This is a modification test"
```

Detection

Wazuh detected integrity changes including:

- File size change
- Modification time change
- MD5 hash change
- SHA1 hash change
- SHA256 hash change

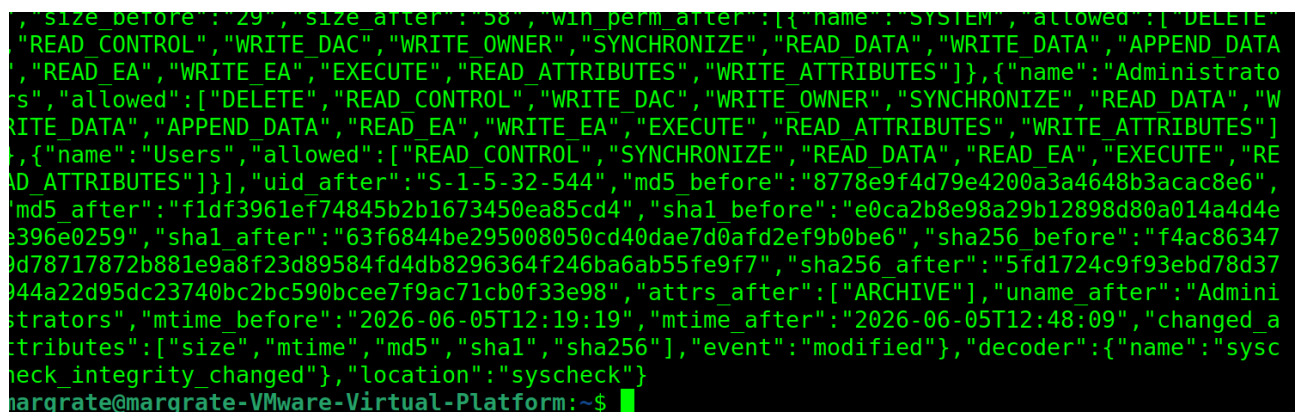
Observed Event:

```
event: modified
```

Decoder:

```
syscheck_integrity_changed
```

```
sudo grep '"id":"002"' /var/ossec/logs/alerts/alerts.json | grep syscheck | tail -20
```



```
, "size_before": "29", "size_after": "58", "win_perm_after": [{"name": "SYSTEM", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Administrators", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Users", "allowed": ["READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "READ_EA", "EXECUTE", "READ_ATTRIBUTES"]}], "uid_after": "S-1-5-32-544", "md5_before": "8778e9f4d79e4200a3a4648b3acac8e6", "md5_after": "f1df3961ef74845b2b1673450ea85cd4", "sha1_before": "e0ca2b8e98a29b12898d80a014a4d4ee396e0259", "sha1_after": "63f6844be295008050cd40dae7d0afd2ef9b0be6", "sha256_before": "f4ac863470d78717872b881e9a8f23d89584fd4db8296364f246ba6ab55fe9f7", "sha256_after": "5fd1724c9f93ebd78d37044a22d95dc23740bc2bc590bcee7f9ac71cb0f33e98", "attrs_after": ["ARCHIVE"], "uname_after": "Administrators", "mtime_before": "2026-06-05T12:19:19", "mtime_after": "2026-06-05T12:48:09", "changed_attributes": ["size", "mtime", "md5", "sha1", "sha256"], "event": "modified", "decoder": {"name": "syscheck_integrity_changed"}, "location": "syscheck"}
```

Figure 11:

File Deletion Test

Action

The file was deleted:

```
Remove-Item C:\FIM-Test\test3.txt
```

Wazuh detection of file deletion event (Rule 553).

```

{"timestamp": "2026-06-05T12:58:03.447+0200", "rule": {"level": 7, "description": "File deleted.", "id": "553", "mitre": {"id": ["T1070.004", "T1485"], "tactic": ["Defense Evasion", "Impact"], "technique": ["File Deletion", "Data Destruction"]}, "firedtimes": 1, "mail": false, "groups": ["ossec", "syscheck", "syscheck_entry_deleted", "syscheck_file"], "pci_dss": ["11.5"], "gpg13": ["4.11"], "gdpr": ["II 5.1.f"], "hipaa": ["164.312.c.1", "164.312.c.2"], "nist_800_53": ["SI.7"], "tsc": ["PII.4", "PII.5", "CC6.1", "CC6.8", "CC7.2", "CC7.3"]}, "agent": {"id": "002", "name": "RainhaStar", "ip": "192.168.116.150"}, "manager": {"name": "margrate-VMware-Virtual-Platform", "id": "1780657083.2377867", "full_label": "CDROM"}, "file": "c:\\fim-test\\test3.txt", "mode": "realtime", "syscheck": {"path": "c:\\fim-test\\test3.txt", "mode": "realtime", "size_after": "58", "win_perm_after": [{"name": "SYSTEM", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Administrators", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Users", "allowed": ["READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "READ_EA", "EXECUTE", "READ_ATTRIBUTES"]}], "uid_after": "S-1-5-32-544", "md5_after": "f1df3961ef74845b2b1673450ea85cd4", "sha1_after": "63f6844be295008050cd40dae7d0afd2ef9b0be6", "sha256_after": "5fd1724c9f93ebd78d37944a22d95dc23740bc2bc590bcee7f9ac71cb0f33e98", "attrs_after": ["ARCHIVE"], "uname_after": "Administrators", "mtime_after": "2026-06-05T12:48:09", "event": "deleted"}, "decoder": {"name": "syscheck_deleted"}, "location": "syscheck"}
margrate@margrate-VMware-Virtual-Platform:~$

```

Figure 12:

Detection

Wazuh generated the following alert:

Rule ID:

553

Description:

File deleted.

Observed Event:

event: deleted

Decoder:

syscheck_deleted

Additional Details:

- Agent: RainhaStar (002)
- Path: C:\FIM-Test\test3.txt
- Monitoring Mode: Realtime

Skills Demonstrated

This project demonstrated the following SOC Analyst skills:

- Wazuh Administration
- File Integrity Monitoring (FIM)
- Windows Endpoint Monitoring
- Linux Log Analysis

- Security Event Investigation
 - Configuration Troubleshooting
 - Alert Validation
 - Threat Detection Workflows
 - Security Operations Center (SOC) Procedures
-

Conclusion

File Integrity Monitoring was configured and validated on the RainhaStar Windows Server 2019 endpoint. The project demonstrated Wazuh's ability to detect file creation, modification, and deletion events in real time.

A configuration error preventing Syscheck monitoring was identified, investigated, and resolved. Following remediation, all File Integrity Monitoring events were successfully detected and logged by the Wazuh Manager.

This project provided practical experience in deploying, troubleshooting, and validating endpoint monitoring controls commonly used in Security

